

WEEK 4 – POWER BI EDA REPORT

SHOPIFY STOCK ANALYSIS

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Department: Data Science and Artificial Intelligence

Tool Used: Microsoft Power BI

Project: Shopify Stock Analysis

01 | PROJECT OVERVIEW

Introduction

In Week 4, I worked on Exploratory Data Analysis (EDA) of Shopify stock data using Microsoft Power BI. The analysis focused on studying stock price movement, trading volume and moving-average trends. I also added interactive filters, a metric selector and a custom tooltip to make the dashboard easier to explore.

Key areas covered:

- Moving Average Analysis
- Price Trend Analysis
- Trading Volume Analysis
- Interactive Date Filtering
- Dynamic Price Metric Selection
- Price Comparison
- Custom Tooltip
- Interactive Dashboard

02 | BUILDING THE MOVING AVERAGE LAYER

2.1 20-Day Simple Moving Average

A **20-Day Simple Moving Average (20 SMA)** was created using DAX to calculate the average Shopify trading volume over a 20-day period.

The measure uses `DATESINPERIOD()` to define the period and `AVERAGE()` to calculate the average volume.

DAX Measure

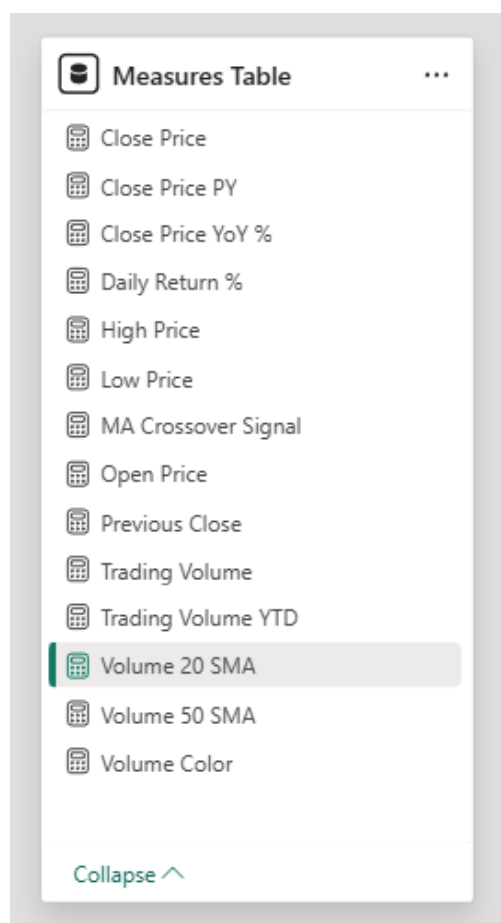
Volume 20 SMA = VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 20

VAR DateInPeriod = DATESINPERIOD('Date Table'[Date],
CurrentDate, -PeriodDays, DAY)

RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]), DateInPeriod)

OUTPUT: A 20-day average-volume measure was created successfully.



2.2 50-Day Simple Moving Average

A **50-Day Simple Moving Average (50 SMA)** was created to calculate the average trading volume over a longer 50-day period.

The same DAX approach was used with a 50-day period. This provides a second moving-average reference for comparison with the 20-day SMA.

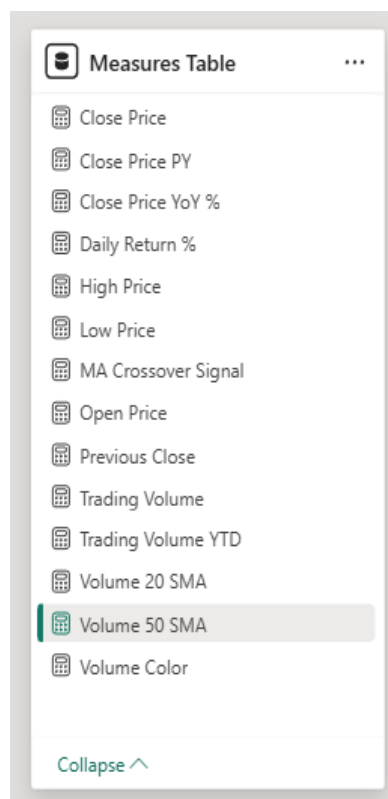
DAX Measure

Volume 50 SMA = VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 50

VAR DateInPeriod = DATESINPERIOD('Date Table'[Date], CurrentDate, - PeriodDays, DAY)

RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]), DateInPeriod)



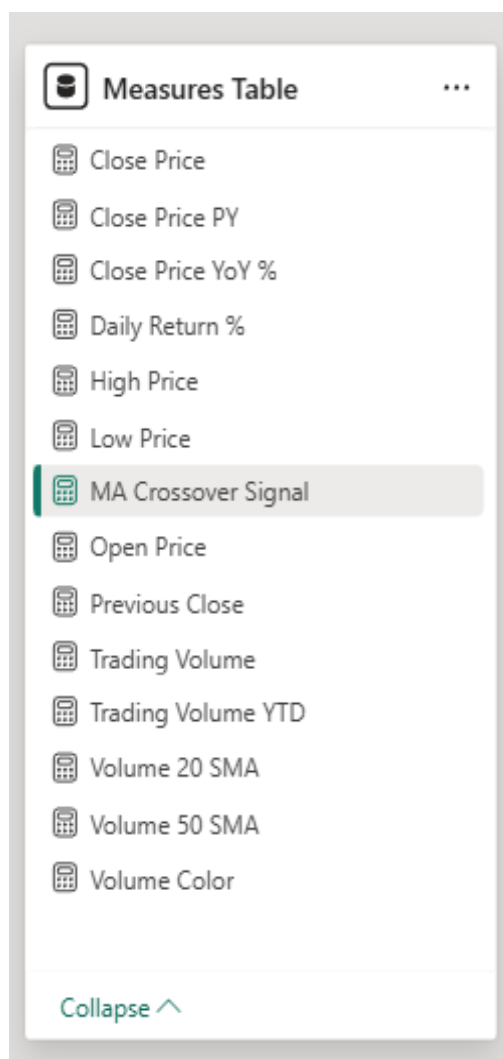
2.3 Moving Average Crossover Signal

A **Moving Average Crossover Signal** was created by comparing the 20 SMA and 50 SMA.

The DAX condition checks the relationship between the two moving averages and returns the corresponding crossover signal.

DAX Measure

```
MA Crossover Signal = IF( ISBLANK([Volume 20 SMA]) ||  
ISBLANK([Volume 50 SMA]), BLANK(), IF([Volume 20 SMA] > [Volume 50  
SMA], "Bullish (20 > 50)", "Bearish (20 < 50)") )
```



03 | READING THE PRICE MOVEMENT

Price Trend & Moving Averages

The main Price Trend & Moving Averages chart was created to display the Close Price along with the 20 SMA and 50 SMA.

Dim_Date[Date] was used on the X-axis, while Close Price, 20 SMA and 50 SMA were added as values. The date axis was configured as Continuous, and the Zoom Slider was enabled for easier timeline exploration.

Visual Elements

X-Axis: Date

Values: Close Price, 20 SMA, 50 SMA

Feature: Zoom Slider



04 | UNDERSTANDING TRADING ACTIVITY

Trading Volume & 20-Day SMA

A Trading Volume & 20-Day SMA chart was created to compare actual trading volume with its 20-day average.

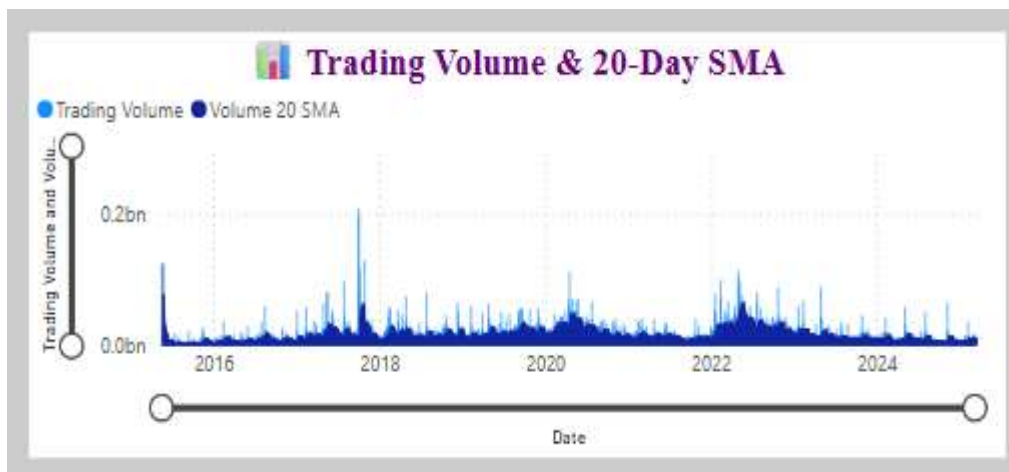
The date field was placed on the X-axis, while Trading Volume and Volume 20 SMA were used as the chart values. This visualization makes it easier to observe changes in trading activity over time.

Visual Elements

X-Axis:Date

Values:Trading Volume,Volume 20 SMA

Feature:Zoom Slider



05 | MAKING THE DASHBOARD INTERACTIVE

5.1 Date Range Slicer

A Date Range Slicer was added using `Dim_Date[Date]`.

The slicer was configured in Between mode so that a specific start date and end date can be selected.

This allows the dashboard to display stock information for a selected date range.



5.2 Relative Date Slicer

A Relative Date Slicer was added to provide quick time-period selections.

The available options are:

- 1 Month
- 3 Months
- 6 Months
- YTD
- 1 Year

This provides a convenient way to explore stock information across different relative periods.



5.3 Metric Selector

A Metric Selector was created using a Field Parameter.

The selector contains:

- Close Price
- High Price
- Low Price
- Open Price

It was created through Modeling → New Parameter → Fields. This allows different price metrics to be selected interactively.

Metric Selector

Metric Selector

☐ Close Price

☐ High Price

☐ Low Price

☐ Open Price

06 | COMPARING PRICE BEHAVIOUR

6.1 High and Low Price Trend

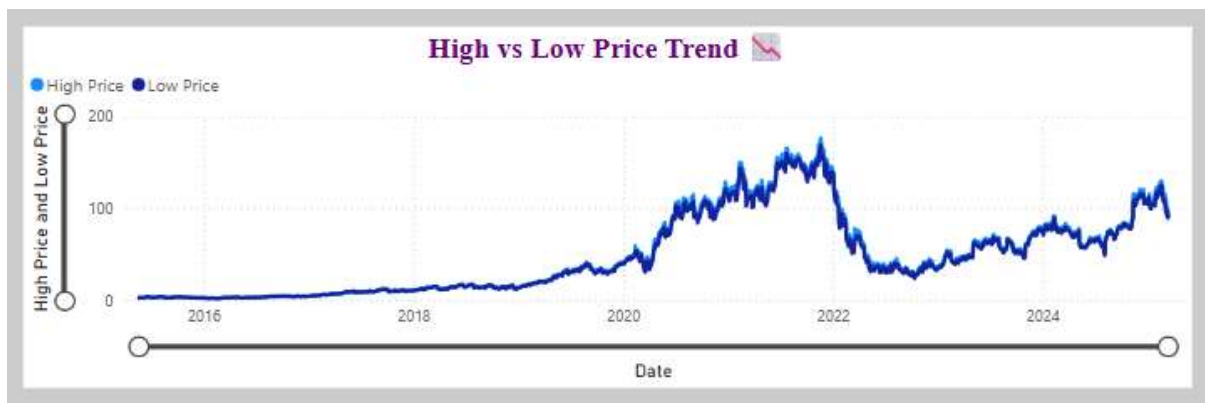
A High and Low Price Trend chart was created to compare the highest and lowest stock prices across different dates.

Visual Elements

X-Axis: Date

Values: High Price, Low Price

Chart: Line Chart



6.2 Open and Close Price Trend

An Open and Close Price Trend chart was created to compare the opening and closing prices of the Shopify stock over time.

Visual Elements

X-Axis: Date

Values: Open Price, Close Price

Chart: Line Chart



07 | ADDING INSTANT INFORMATION WITH TOOLTIP

Custom Tooltip

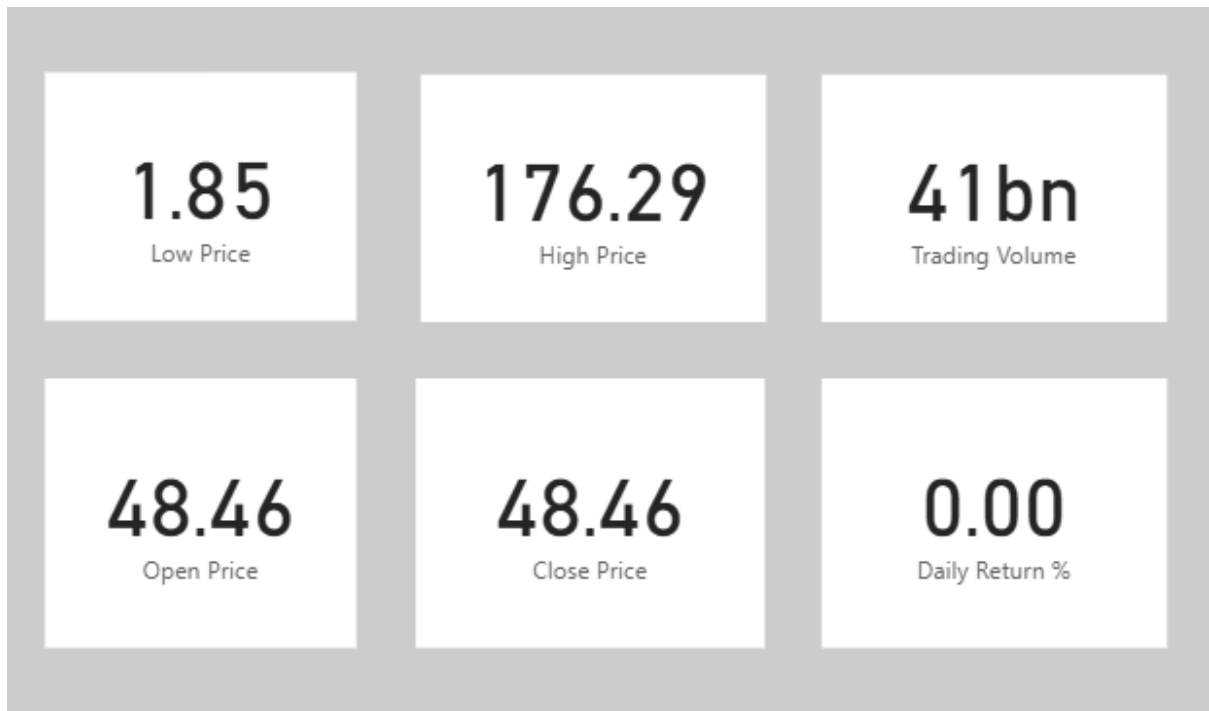
A separate report page named Stock_Tooltip was created.

The page was configured as a tooltip by enabling Allow as tooltip and setting the canvas type to Tooltip.

The tooltip contains detailed information such as:

- Open Price
- High Price
- Low Price
- Close Price
- Volume
- Daily Change %

The tooltip page was then connected to the main price chart so that additional information can be displayed when interacting with the chart.



08 | FINAL INTERACTIVE DASHBOARD

The final Shopify Stock Analysis Dashboard combines the major analysis components into one interactive report page.

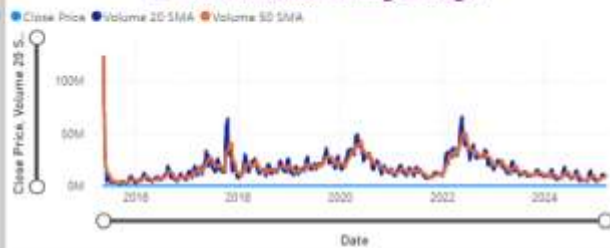
Dashboard Components

- Price Trend & Moving Averages
- Trading Volume & 20-Day SMA
- Date Range Slicer
- Relative Date Slicer
- Metric Selector
- High and Low Price Trend
- Open and Close Price Trend

The combination of these components provides an interactive view of Shopify stock price and trading-volume information.

Shopify Stock Analysis Dashboard

Price Trend & Moving Averages



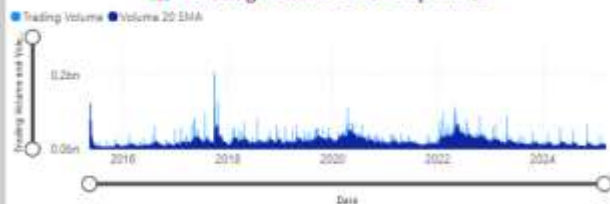
Date Range Slicer



Relative Date Slicer



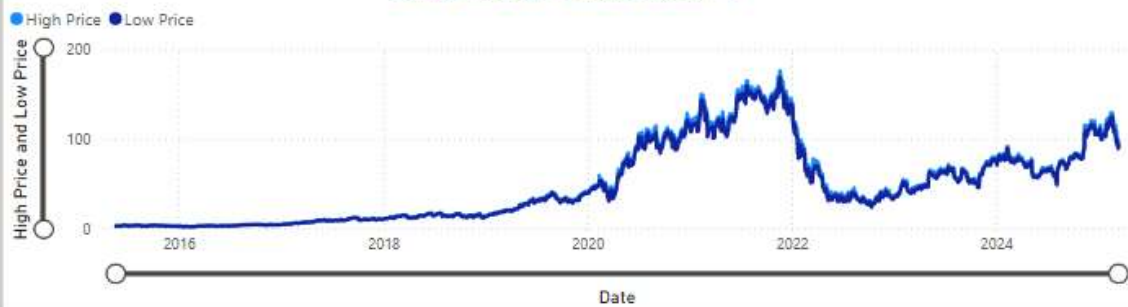
Trading Volume & 20-Day SMA



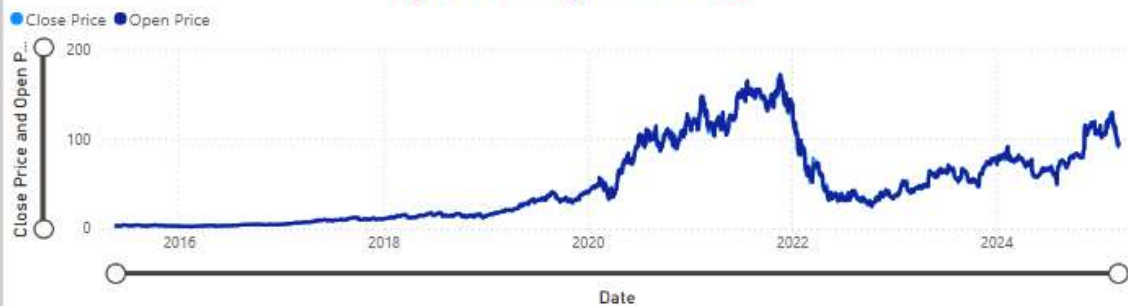
Metric Selector

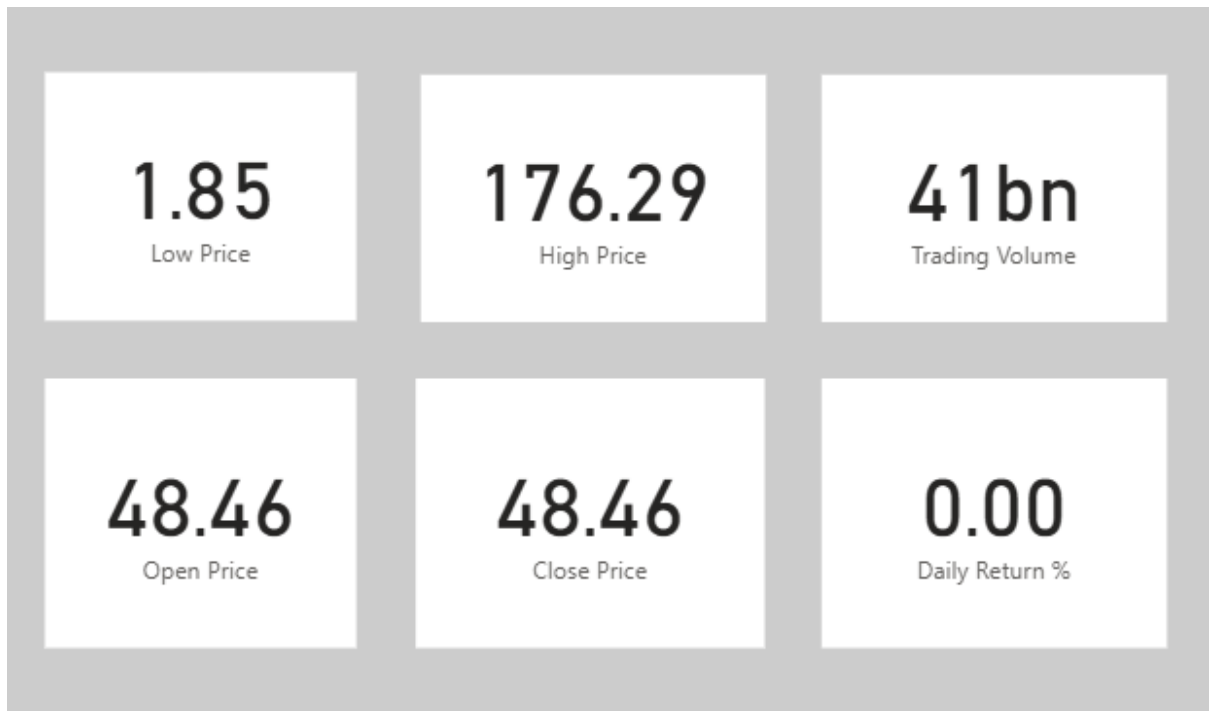


High vs Low Price Trend



Open vs Close price Trend





09 | KEY LEARNING OUTCOMES

Through this task, I gained practical experience in:

- Creating DAX measures
- Calculating 20-Day and 50-Day SMA
- Working with moving-average comparisons
- Building stock price and volume visualizations
- Creating interactive slicers
- Using Field Parameters
- Creating custom tooltips
- Designing an interactive Power BI dashboard

10 | CONCLUSION

The Week 4 Power BI EDA task helped me understand how stock data can be explored through moving averages, price trends, trading volume and interactive dashboard features.

By combining DAX calculations with different Power BI visualizations and interactive controls, I created a Shopify Stock Analysis Dashboard that presents the stock data in a clear and organized manner.

